

HARSHAL KOLHE

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EDUCATION

Visvesvaraya National Institute of Technology, India

Bachelor of Technology in Chemical Engineering, Minor in Internet of Things

Aug' 23 – May' 27

CGPA: [8.66/10.0](#)

EXPERIENCE

Indian Institute of Technology, Roorkee

May' 25 – Jul' 25

Summer Research Intern | [\[Certificate\]](#)

- Developed a reinforcement learning-based control system for a quadcopter carrying a suspended payload, with active suppression of payload-induced oscillations during flight.
- Built a high-frequency data collection and processing pipeline that synchronized multi-source flight and payload unit data streams for reinforcement learning policy validation.
- Built a real-time autonomy stack that streamed 20+ state variables from CubeOrange at 100 Hz via MAVLink, performed onboard RL inference, and generated flight control commands for autonomous stabilization.

PROJECTS

Dual UAV Scan and Spray Architecture for Precision Agriculture | [\[GitHub\]](#)

Jan' 26 – Apr ' 26

- Developed a real-time onboard crop disease detection pipeline using OpenCV and HSV-based image segmentation, under varying outdoor lighting conditions through dual-mask filtering and ND-filter-assisted preprocessing.
- Built a dual-nozzle spraying system with real-time nozzle diameter selection based on detected plant clusters; developed the Raspberry Pi-based control algorithm and designed, fabricated, and assembled the control PCB for automated operation.
- Designed and fabricated a custom carbon-fiber quadcopter frame capable of carrying a 10 kg fluid payload.

Multiagent Hybrid Disaster Rescue Drones | [\[GitHub\]](#)

Jul' 25 – Jan' 26

- Developed a two-drone system that searches in a lawnmower pattern, detects humans, and diverts to deliver relief packages.
- Fine-tuned and deployed a YOLO object detection model for aerial imagery, achieving 90% human detection accuracy at 8m, and integrated it into an end-to-end autonomous localisation pipeline that generated target GPS coordinates and performed dynamic UAV redirection with cluster-based filtering.
- Developed the core telemetry processing layer for a multi-UAV visualisation platform, combining MAVLink data streams and enabling real-time tracking, status monitoring, and mission visualisation.

Design and Development of Fixed Wing UAV | [\[GitHub\]](#)

May' 24 – Oct' 24

- Constructed a custom fixed-wing UAV, with a 1.25m wingspan, 5:1 aspect ratio, and 25 cm chord for aerodynamic performance.
- Conducted aerodynamic analysis of various airfoils in XFLR5 to achieve a target 2D C_L of 1.0 at 4–5° angle of attack and made the aircraft's CAD model based on aircraft design principles.

PUBLICATIONS

- R. Deshmukh, **H. Kolhe**, S. Biswas, and S. S. Chiddarwar, "Hybrid Controller Development for a VTOL Using PID, LQR and MPC," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- S. Kshirsagar, **H. Kolhe**, R. Deshmukh, K. Ayush, S. Dhalwar, S. Biswas, M. Limaye, and S. S. Chiddarwar, "A Multi-Agent Drone System for Real-Time Victim Detection and Aid Delivery," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- A. Sharma, M. Ravikiran, S. Chakrabarty, R. Saluja, M. Limaye, **H. Kolhe**, and S. Samiron, "Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs," 2025 IEEE International Conference on Big Data (BigData), Macau, China, 2025, pp. 7278-7286, doi: [10.1109/BigData66926.2025.11401432](https://doi.org/10.1109/BigData66926.2025.11401432).

SKILLS

Languages: C, C++, Python, Java

Software and Tools: ArduPilot, QGroundControl, Mission Planner, Linux, ROS2, MATLAB

Libraries: OpenCV, Ultralytics (YOLOv8), NumPy, Pandas, Matplotlib, DroneKit, MAVSDK, MAVlink

EXTRACURRICULAR ACTIVITIES

- Secured the [Best Paper Presentation Award](#) at the ICAMMS Conference for presenting research work on autonomous systems.
- Secured the Ignite Prize in **NIDAR**, a national-level UAV autonomy Competition, placing among the top 5 out of 100+ teams.
- Secured [First Runner-up](#) prize at the Drone Helix Competition, SVNIT Surat.
- Vice Chairman of IvLabs, Robotics and Artificial Intelligence Lab at VNIT, Nagpur.